

Dineth Perera

Electrical & Electronic Engineering Undergraduate | Computer Vision | Signal Processing | Remote Sensing

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PROFESSIONAL SUMMARY

Third-year B.Sc. Engineering undergraduate in Electrical and Electronic Engineering at the University of Peradeniya (GPA: 3.74/4.00), with research experience in remote-sensing computer vision, visual state-space models, binary change detection, semantic segmentation, signal processing. Seeking research internship opportunities in computer vision, remote sensing, generative AI, and medical imaging

EDUCATION

B.Sc. Engineering (Hons) in Electrical and Electronic Engineering

2023 - Present

University of Peradeniya, Sri Lanka

- Current GPA: 3.74/4.00

G.C.E. Advanced Level - Physical Science Stream

2021

Sri Lanka

- National Rank: 68 | District Rank: 4 | Z-score: 2.62

PUBLICATIONS AND MANUSCRIPTS

A Controlled Benchmark of Visual State-Space Backbones with Domain-Shift and Boundary Analysis for Remote-Sensing Segmentation

IGARSS 2026

Accepted, IEEE International Geoscience and Remote Sensing Symposium
| [arXiv:2604.18721v1](#)

Nichula Wasalathilaka, **Dineth Perera**, Oshadha Samarakoon, Buddhi Wijenayake, Roshan Godaliyadda, Vijitha Herath, Parakrama Ekanayake.

- Contribution: experiments, benchmarking, analysis, and manuscript preparation; compared visual SSMs with controlled CNN/Transformer baselines on LoveDA and ISPRS Potsdam under a fixed lightweight decoder.

MambaRefine-CD: MambaVision with Region-Boundary Temporal Refinement for Remote-Sensing Change Detection

MERCon 2026

Manuscript / Under Review

Dineth Perera, Thaaariq Firdous, Oshadha Samarakoon, Buddhi Wijenayake, Roshan Godaliyadda, Parakrama Ekanayake, Vijitha Herath.

- Contribution: developing a MambaVision-based binary change detection architecture with differential region-boundary interaction, ARF-FPN multiscale decoding, and bounded boundary residual refinement.

Task-Aware Evaluation of Generative Models for Ultra-Long Vibration Signals in Predictive Maintenance

IRAI 2026

Manuscript / Under Review

Oshadha Samarakoon, **Dineth Perera**, Kavishka Galagoda, Thaaariq Firdous, Fikry Nuhuman.

- Contribution: experiments, methodology development, evaluation design, result analysis, and manuscript preparation for generative PdM benchmarking.

ENGINEERING PROJECTS

Real-Time Elephant Detection Using Infrasound Analysis and Machine Learning

2025

Signal processing, ML, sensing, embedded systems

- Developing an infrasound-based elephant detection approach using low-frequency acoustic signals in the 5-35 Hz range, which can propagate over long distances and remain below normal human hearing thresholds.
- Focus areas include signal acquisition, preprocessing, spectral feature extraction, noise robustness, machine-learning classification, and eventual embedded/field deployment.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, LaTeX, Git, Linux

Machine Learning / Computer Vision: PyTorch, OpenCV, TensorFlow, semantic segmentation, CNN/Transformer baselines, visual state-space models.

Signal Processing / Time Series: DSP, FFT/PSD analysis, filtering, spectral diagnostics, vibration and infrasound analysis, predictive-maintenance evaluation

Embedded Systems / Hardware: Arduino/AVR, ESP32, Raspberry Pi, Jetson, real-time monitoring systems

HONORS AND AWARDS

- P.D.S. Kularathne and N.B.M. Mediweka Memorial Prizes for the best student selected to the Faculty of Engineering in 2021.
- S.J. Munasinghe Memorial Prize for the student who obtained the highest Z-score at the G.C.E. Advanced Level Examination in 2021.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Central Link Toastmasters Club

2023 - 2026

Member

- Developed technical communication, public speaking, and structured presentation skills through club participation.